

Gene Pulser® Electroprotocol

Cell Type	Bacterial, gram negative.	Molecules Electroporated	DNA: plasmid or ligation mixture; 3 to 100 kB
Species Used	<i>Agrobacterium</i> , unspecified species		

Before the Pulse

Cell Growth Medium	LB	Growth Phase at Harvest	O.D. (600) =0.7-1.0
		Pre-pulse Incubation	none

Wash Solution 10% glycerol,1mM HEPES, pH 7.0; twice at equal volume (1x)

The Pulse **Instruments Used** Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium 10% glycerol,1mM HEPES, pH 7.0

Cuvette Gap 0.2 cm

Voltage 2.5 kV

Cell Density 1x10 (11) cells / ml

Volume of Cells 30-40 µl

Field Strength 12.5 kV/cm

DNA Concentration 100 ng / 3 µl

DNA Resuspension Buffer SOB or SOC

Capacitor 25 µF

Volume of DNA 3 µl

Resistor 200 Ω (Pulse Controller)

After the Pulse

Time Constant 3.8 msec

Outgrowth Medium SOB or SOC

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Comments: The *vir C* and *vir D* operons of the *Agrobacterium* Ti plasmid are regulated by the *ros* chromosomal gene: analysis of the cloned *ros* gene. Michael B. Cooley, Maria R. D'Sowza and Clarence I. Kado (1991) *J. Bacteriology* **173**(8). **SOB:** 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 20 mM glucose. **SOC:** 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, 10 mM MgSO₄,20 mM glucose. **LB:** 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

Outgrowth Temperature	25 °C
Length of Incubation	2hrs+
Selection Method or Assay Used	unspecified antibiotic
Electroporation Efficiency	5x10 (8) to 1x 10 (9) transformants / µg DNA
Per Cent Survival	10%

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Survey Number
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